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Equivalent or value less than power of 2 is : 64

(1) 65 - 64 = 1

(2) 1 - 1= 0

Power's of 2	64	32	16	8	4	2	1
Binary Number	1	0	0	0	0	0	1

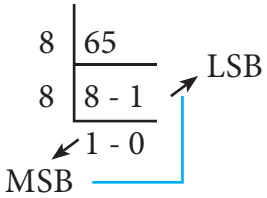
$65_{10} = (1000001)_2$

2.4.2 Decimal to Octal Conversion

To convert Decimal to Octal, "**Repeated Division by 8**" method can be used. The method is the same we have learnt in 2.4.1, but in this method, we have to divide the given number by 8.

Example

Convert $(65)_{10}$ into its equivalent Octal number



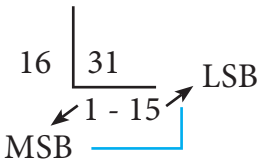
$(65)_{10} = (1\ 0\ 1)_8$

2.4.3 Decimal to Hexadecimal Conversion

To convert Decimal to Hexadecimal, "**Repeated division by 16**" method can be used. The method is the same as we have learnt in 2.4.1, but in this method, we have to divide the given number by 16.

Example

Convert $(31)_{10}$ into its equivalent hexadecimal number.



$(16)_{10} = (1F)_{16}$(Refer Table 2.2 F=15)

2.4.4 Conversion of fractional Decimal to Binary

The method of **repeated multiplication by 2** has to be used to convert such kind of decimal fractions.

The steps involved in the method of **repeated multiplication by 2**:

Step 1: Multiply the decimal fraction by 2 and note the integer part. The integer part is either 0 or 1.

Step 2: Discard the integer part of the previous product. Multiply the fractional part of the previous product by 2. Repeat Step 1 until the same fraction repeats or terminates (0).

Step 3: The resulting integer part forms a sequence of 0s and 1s that become the binary equivalent of decimal fraction.

Step 4: The final answer is to be written from first integer part obtained till the last integer part obtained.